# QUARTERLY OPERATIONS REVIEW

Safety, Equipment Health & Supply Chain Performance

Presented to the Senior Management Team

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Week 8

EXECUTIVE SUMMARY

# Bottom Line Up Front

**Demand forecasting and inventory optimization work delivered this quarter cuts projected stockout frequency by 3x and identifies a $38K/week least-cost distribution plan — while equipment health monitoring continues to reduce unplanned downtime risk.**

FORECAST ACCURACY

**94.2%**

*(5.79% MAPE, 45-day horizon)*

STOCKOUT RISK REDUCTION

**3x lower**

*1.5% -> 0.5% with safety stock*

WEEKLY DISTRIBUTION COST

**$38,180**

*optimized via linear programming*

EXECUTIVE SUMMARY

# What This Means & What We're Asking For

## WHERE WE STAND

- Demand forecasting is now accurate enough (5.79% MAPE) to size inventory buffers with confidence, not guesswork.
- Equipment health monitoring (Weeks 6-7) continues to catch failure risk before it becomes downtime.
- Distribution costs are now optimized mathematically, not by habit or plant-of-convenience.

## WHAT WE'RE ASKING OF THIS ROOM

- Approve the reorder-point policy change (Slide 6) for the next replenishment cycle.
- Sign off on the 3 strategic recommendations on Slide 8, including budget for the highest-ROI item.
- 10 minutes for Q&A at the end to pressure-test assumptions before rollout.

**Agenda: Safety & Equipment Health -> Supply Chain Performance -> Strategic Recommendations -> Q&A**

# Predictive Maintenance Model

24 pumps, 4 depots, 28 days of sensor data (vibration, temperature, pressure, acoustic, motor current) — engineered into rolling health features and two predictive models.

| | |
|---|---|
| **VALIDATED DRIVERS** | Night-shift vibration runs ~4% higher than day (Welch's t=-2.74, p=0.006). Depot temperature differs significantly (ANOVA F=17.47, p<0.001): West runs coolest at 63.3°C, South hottest at 65.8°C. |
| **FAILURE CLASSIFIER** | Logistic regression flags “failure within 7 days” at 92.6% accuracy, 0.92 AUC on held-out pumps — catches 139 of 166 true failures (84% recall). |
| **REMAINING-LIFE MODEL** | Linear regression estimates hours-to-failure with ~5-day typical error (MAE 118h, $R^2$=0.46) — precise enough to rank which flagged pump to inspect first. |

**92.6% / 0.92**

Failure-classifier accuracy / AUC

**~5 days**

Remaining-life estimate error (MAE)

**24 pumps, 4 depots**

Fleet currently monitored

# Safety Incident & Compliance Analysis

304 incidents across 5 U.S. depot/plant sites, Jan–Jun 2026, from the HSE dashboard's live dataset.

## Root-cause themes identified

- Fall from Height, Vehicle Interaction, and Chemical Exposure are only ~24% of incident volume but account for 62% of potential-SIF events.
- Critical-control verification lags sharply at Depot B (77.5%) vs. Terminal E (97.1%) — a ~20-point controls-execution gap, not a training gap (training compliance is ~89–92% across every site).
- Incident count held roughly flat quarter-over-quarter (154 → 150), while average severity ticked up (2.25 → 2.35).

## Where this leaves us

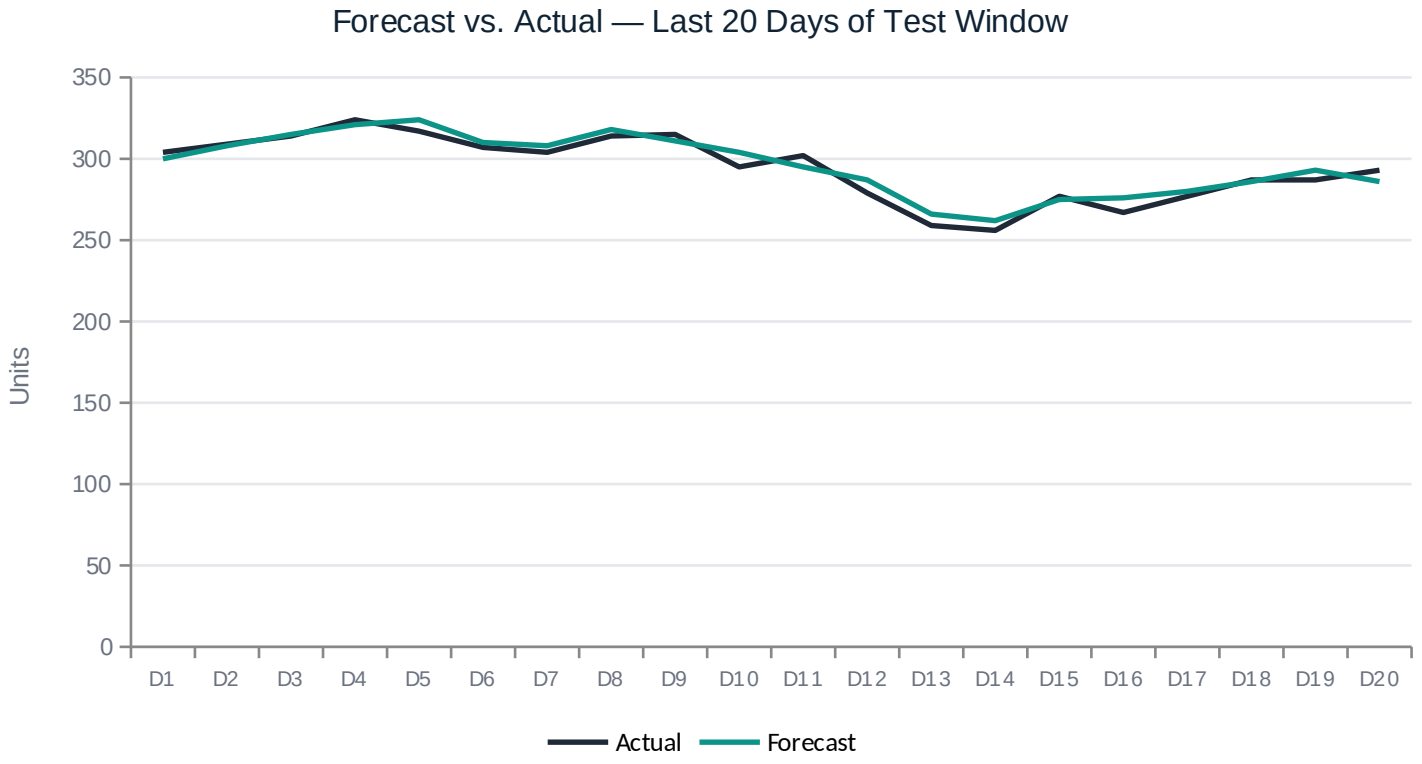
304 total incidents | 168 recordable (55.3%)
83 lost-time | 71 near-miss | 65 potential SIF
Critical-control verification: 86.8% fleet avg
Inspection completion: 84.1% fleet avg
Training compliance: 90.9% fleet avg

Full detail in the live HSE dashboard — flagging here only what's relevant to this quarter's operational decisions.

***Signal worth acting on: the three highest-severity incident types (Fall from Height, Vehicle Interaction, Chemical Exposure) drive 62% of SIF-potential risk despite only 24% of volume — targeted controls there outweigh broad, low-severity fixes.***

# Demand Forecast: Accuracy & Horizon

A Prophet time-series model — trend, weekly/yearly seasonality, plus promotion, holiday, and weather signals — forecasts daily demand 45 days out.




**5.79%**

MAPE (test set)

**22.4 units**

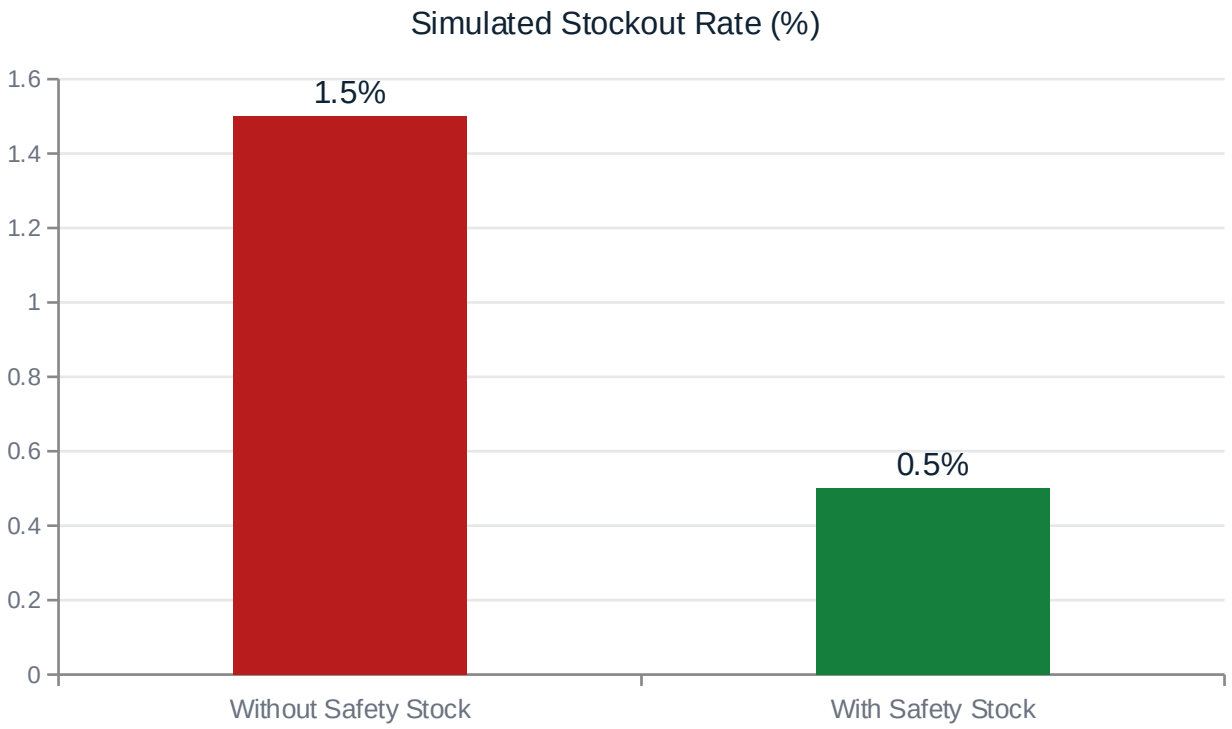
RMSE (test set)

**45 days**

Forecast horizon

# Inventory Policy: Safety Stock Impact

Safety stock is sized off forecast error (not raw demand variance) at a 95% service level, 7-day lead time. We simulated 200 Monte Carlo runs to test stockout frequency with and without it.




**Safety Stock**

**98 units**

**Reorder Point**

**2,199 units**

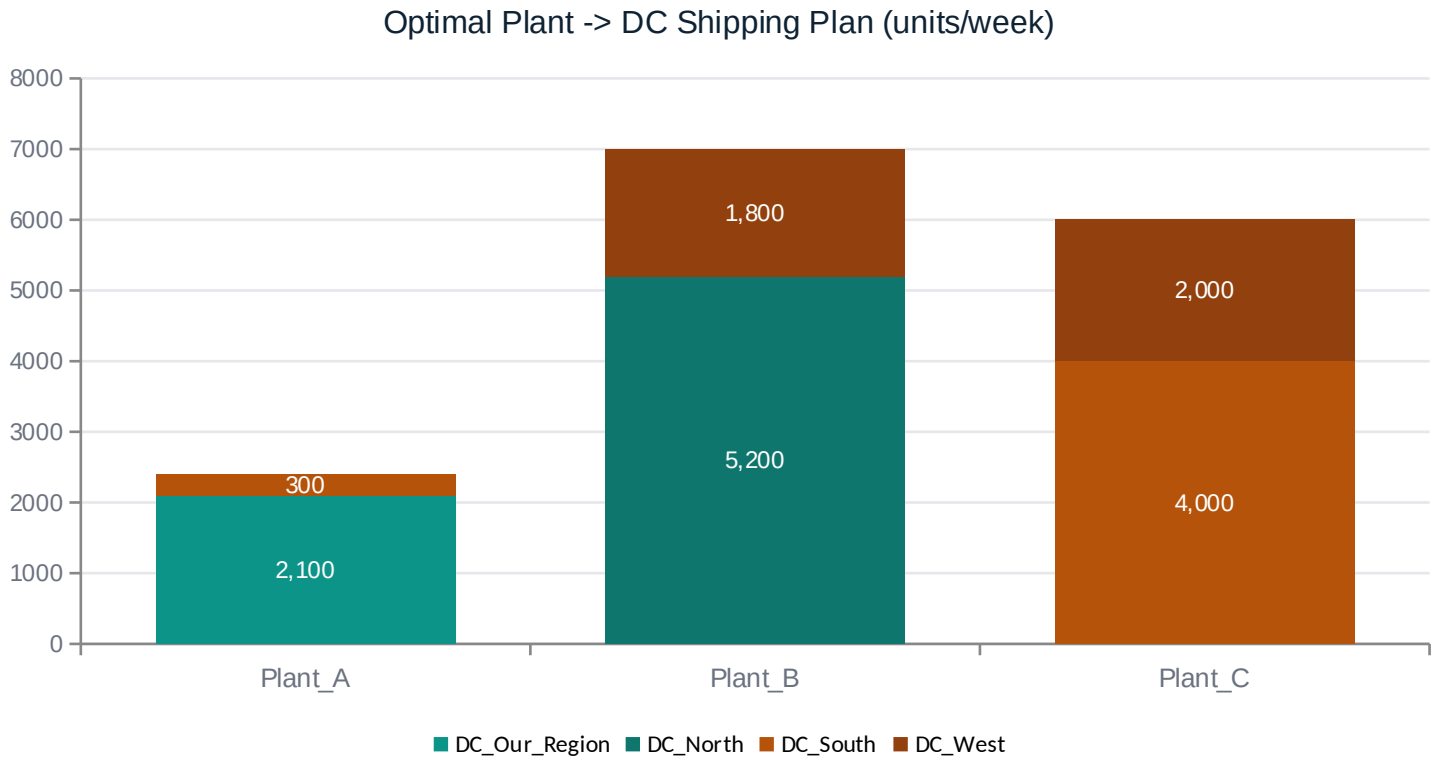
**Service Level Target**

**95%**

**Recommendation: adopt the 2,199-unit reorder point for next cycle.**

# Distribution Network: Least-Cost Plan

A linear program (PuLP) assigns weekly shipments from 3 plants to 4 regional DCs, minimizing total shipping cost subject to plant capacity and forecasted DC demand.




**$38,180**

Optimal weekly shipping cost

**15,400 units**

Total demand satisfied

**Optimal**

Solver status (CBC)

STRATEGIC RECOMMENDATIONS

# Three Actions, With ROI

1

## Adopt the 2,199-unit reorder point policy

Replace the current static reorder rule with the forecast-error-based reorder point starting next replenishment cycle.

***Est. ROI: ~$137/mo avoided lost-margin vs. ~$65/mo added holding cost (~2.1x) — assumptions below****

2

## Re-route plant volume per the LP plan

Split DC_South and DC_West across Plant A/B/C exactly as the optimizer assigns, instead of an even, cost-unaware split across all three plants.

***Est. ROI: $19,320/week (33.6%) lower shipping cost vs. an even-split baseline — ~$1.0M/year***

3

## Cut the PdM miss-rate & audit Depot B's controls

Tighten sensor coverage on the existing 24-pump fleet to shrink the classifier's 27-failure miss rate; run a critical-control audit at Depot B (77.5% verified vs. 97.1% at Terminal E).

***Est. cost: ~$3,600 sensor upgrade + ~$3,000 audit (~$6,600 one-time)* — payback: avoided cost of one prevented failure/incident, TBC with Finance***

**Illustrative estimates using standard planning assumptions ($40/unit landed cost, 20% annual carrying rate, $15/unit gross margin, 10% unmet demand per stockout day, $75/hr loaded audit rate) — replace with actual finance figures before final budget approval.*

# Discussion & Q&A

Thank you — happy to take questions.